

Arundhati Vasisht

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EDUCATION

University of Massachusetts, Amherst

BS, Computer Science; GPA: 3.83; Chancellor's Award, Dean's List Honoree

Amherst, MA

Expected Fall 2028

SKILLS

Languages: Python, Java, TypeScript, JavaScript, HTML/CSS

Frameworks & Cloud: React.js, Node.js, Express, Flask, TensorFlow/Keras, AWS (Bedrock, Kendra), MongoDB, REST APIs, Git

Data & AI: RAG pipelines, LLM integration, statistical analysis, data modeling, visualization, hypothesis testing

EXPERIENCE

Massachusetts Executive Office of Health & Human Services (EOHHS)

Boston, MA

AI Software Engineer

Spring to Summer 2026

- Built an AI-powered regulatory chatbot for Massachusetts DPH's statewide public health platform (Metrik), targeting 85%+ accuracy across 351 local health departments and 5 inspection domains.
- Architected a RAG pipeline on AWS Bedrock with Amazon Kendra, grounding all responses in official Massachusetts CMR documents to eliminate hallucination risk for court-level enforcement decisions.
- Conducted stakeholder interviews with DPH to define KPI benchmarks, reducing inspector document search time from 2 to 10 minutes per query to near-instantaneous AI responses.
- Designed audit logging, inspector feedback, and flagging systems to continuously monitor response accuracy, completeness, and recall statewide.

Indian Institute of Science (IISc)

Bangalore, India

Machine Learning Research Intern

Summer 2025

- Conducted a comparative study of 6 ML/DL pipelines (SVM, PCA-SVM, CNN, XGBoost, PCA-XGBoost, CNN-XGBoost) for medical image classification using 8,700+ chest X-ray and brain MRI images.
- Implemented VGG16 transfer learning and hybrid CNN-XGBoost pipelines, achieving up to 96% test accuracy on brain MRI and 95% on chest X-ray classification, outperforming standalone SVM and CNN models.
- Evaluated model robustness across multiple splits using accuracy, F1, ROC-AUC, and log-loss; collaborated with 11 researchers to identify hybrid deep learning as the most reliable approach for limited-data medical imaging.

Minutemen Alternative Investment Fund (MAIF)

Amherst, MA

Quant Arm, Junior Analyst

Spring 2026

- Built a multi-agent NLP pipeline semantically matching prediction-market contracts across 4 exchanges (Kalshi, Polymarket, Manifold, Metaculus) to find deterministic arbitrage pairs.
- Designed a pipeline embedding market questions into a Qdrant vector DB, clustering via KNN, and using an LLM (Groq/Llama) to classify semantic relationships across 5 relation types, scoring price mismatches.
- Built a live polling service that re-fetches quotes for confirmed pairs and alerts when profit edge exceeds a configurable threshold.

PROJECTS

FingerSolve | AI-Powered Hand Gesture Recognition Platform

Spring 2025

- Built a real-time gesture-recognition learning tool for touch-free math quizzes, improving engagement for children and users with disabilities by 40% during pilot testing.
- Engineered a React/Flask platform using MediaPipe Hands + TensorFlow/Keras, achieving 94% gesture-classification accuracy across 12 gestures; reduced user input errors by 30% with confidence scoring.
- Deployed a live prototype that earned 2nd place at the MassAI ML Conference and attracted interest from 50+ educators.

MedEcho | AI-Powered Medical Results Translator

Spring 2026

- Built a 3-portal system where a DenseNet-121 CNN classifies chest X-ray findings and GradCAM highlights the driving region, turning scans into plain-language explanations in the patient's own language.
- Used Gemini to generate patient scripts and ElevenLabs + HeyGen's LiveAvatar to deliver diagnoses out loud, face-to-face, in 4 languages, before and after physician review.
- Added clinical-trial matching using the ClinicalTrials.gov API to score patient eligibility; built the full Flask/React prototype in 24 hours.

EcoForecast | Full-Stack Sustainability Analytics Platform

Spring 2026

- Built a full-stack platform simulating 20-year financial ROI and carbon impact of green investments for small businesses using custom TypeScript/Node.js financial models.
- Developed backend APIs and MongoDB workflows to process quarterly utility data and generate cost projections, break-even analysis, and carbon reduction estimates.
- Integrated a React dashboard with 4+ visualizations and a LLaMA 3.3-powered AI advisor for context-aware sustainability insights.